IT4010 Research Project – 2025 July

Topic Assessment Form

Project ID : 25-26J-127

1. Topic (12 words max)

LegalVision: A Knowledge-Driven AI Framework for Legal Contract Understanding and Trust Assessment

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Data Science (DS)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

In the evolving digital landscape, legal systems remain foundational across many sectors but are often hindered by inefficiencies, complexity, and inaccessibility. Legal professionals and laypersons alike face challenges when engaging with legal documents due to their volume, ambiguity, and specialized jargon. Simultaneously, the emergence of Large Language Models (LLMs), such as GPT-4, offers transformative potential to streamline contract-related processes. However, legal-specific applications of LLMs are still immature, with major limitations in explainability, data transparency, localization, and ethical use. This research addresses these gaps by proposing a domain-specific, interpretable, and trustworthy AI framework for Contract Law, tailored particularly to the Sri Lankan legal context.

Current summarization tools fail to handle legal texts effectively. They often generate overly generic summaries that do not align with user intent or legal context[2]. Legal documents are typically dense, highly structured, and clause-based, requiring fine-grained understanding. In broader legal NLP research, studies like Chalkidis et al. have shown that general-purpose models underperform on legal texts due to the complexity and specificity of legal language [1]. However, there remains a clear research gap in tools that can provide legal summaries using clear and simplified terms, adapt the summary to the user's specific legal query or context, and represent key elements of the summary visually. These limitations are particularly evident in countries like Sri Lanka, where users often require more accessible, context-aware, and interpretable summaries to navigate legal documents effectively.

Legal professionals demand not only accurate outputs but also traceable and transparent reasoning. However, most large language models (LLMs) operate as “black boxes,” offering conclusions without explaining the reasoning behind them — a limitation that is particularly problematic in high-stakes legal settings. As highlighted by Zhong et al., existing AI systems lack the ability to provide structured, interpretable reasoning processes that align with legal thinking. Although some prior work has explored rule-based or logic-driven legal reasoning, there is a notable gap in integrating such interpretability into LLM-generated outputs. This lack of explainability reduces trust, auditability, and legal accountability in AI-generated decisions within the legal domain[1].

Legal knowledge is often locked in unstructured formats such as court documents, contracts, and gazette publications, making it difficult to search, analyze, or connect legal concepts effectively. Traditional legal knowledge graphs (KGs) are typically static, limited in scope, and unable to reflect the evolving nature of legal systems. Research by Grover et al. highlights the lack of dynamic, interconnected legal KGs, especially in low-resource and underrepresented legal jurisdictions. This presents a significant gap in enabling automated understanding, traceability, and semantic connectivity across legal content[1].

Legal AI systems must be held to higher standards of fairness and transparency due to the sensitive and high-stakes nature of legal decision-making. However, most existing bias detection tools are designed for generic content domains, such as social media or news, and are not suitable for the legal context. These tools often fail to detect domain-specific forms of bias, such as procedural or sentencing bias, and do not account for the complexities of legal language, precedent, or statutory interpretation. Furthermore, they typically lack explainability, adaptability to evolving legal norms, and comprehensive trust assessment mechanisms. This reveals a critical research gap in creating legal-domain-specific frameworks that can identify and evaluate bias and trustworthiness in a context-aware and legally grounded manner[1].

Significant gaps remain in legal AI systems related to clarity, reasoning transparency, knowledge structuring, and fairness evaluation. Existing tools lack the legal-specific focus needed to address these critical challenges effectively.

References

1. Lai, J., Gan, W., Wu, J., & Qi, Z. P. S. (2024). Large language models in law: A survey. *AI Open, 5*, 181–196. https://doi.org/10.1016/j.aiopen.2024.09.002
2. Zhong, H., Zhang, C., et al. (2020). How does NLP benefit the legal system: A summary of recent progress. *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, 5218–5230. https://aclanthology.org/2020.acl-main.466

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

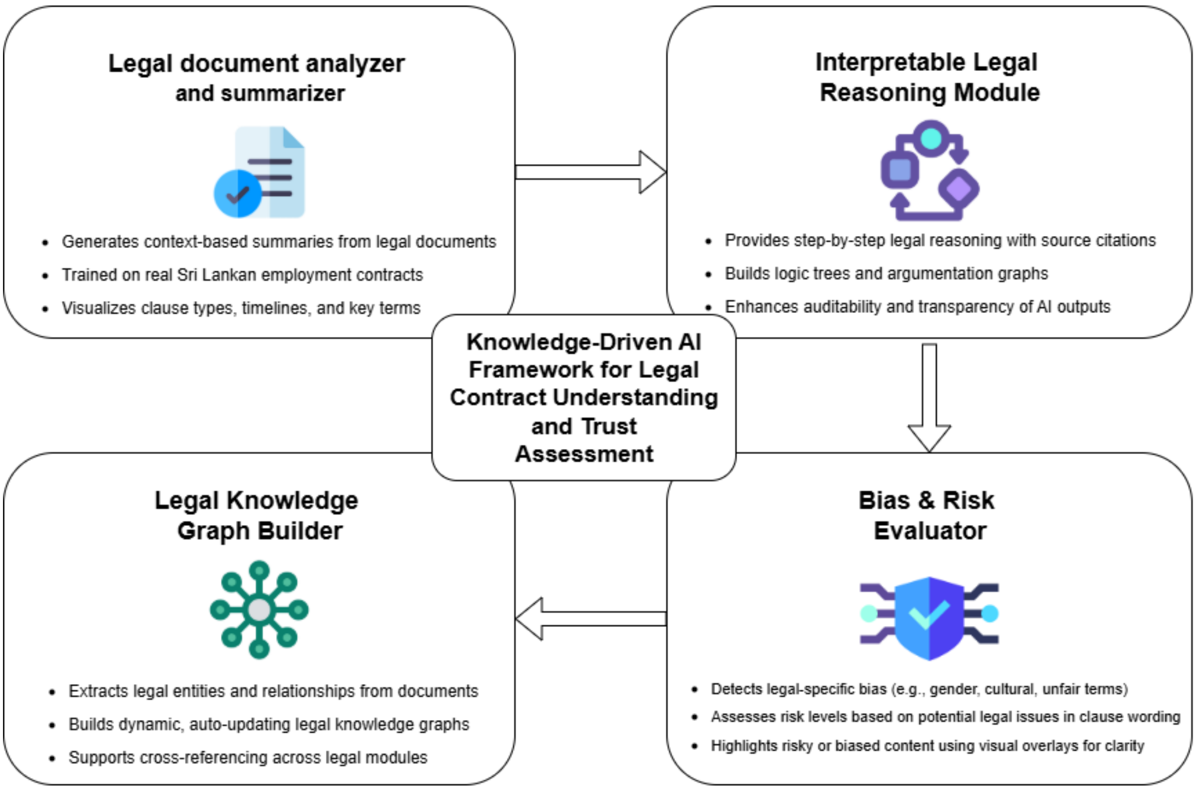
The proposed system is an AI-based legal framework designed to make the analysis of employment agreements more interpretable, accessible, and trustworthy. It combines advanced techniques in Natural Language Processing (NLP), lightweight legal knowledge representation, and bias/risk detection. All components are tailored to suit the structure and regulatory context of Sri Lankan labor law, with the goal of supporting legal professionals and HR personnel in reviewing, understanding, and evaluating the content of employment contracts.

The system includes a legal document analyzer that uses Retrieval-Augmented Generation (RAG) and transformer models such as LegalBERT, mBERT, or Gemini/GPT to generate query-specific summaries from employment agreements. Documents are segmented at the clause level using rule-based or model-driven methods, while visual analytics (via libraries like D3.js or Plotly) highlight timelines, clause types, and structural gaps—such as missing leave, termination, or benefits clauses—using a Sri Lanka-specific contract clause checklist.

To ensure transparency in reasoning, the framework uses prompt chaining with LLMs to simulate legal reasoning through natural language. It also applies graph-based techniques (e.g., Neo4j or simplified clause-linking methods) to show relationships between rights, obligations, and enforcement pathways within the contract.

A dedicated bias and risk evaluator, tailored to employment clauses, identifies bias types such as gendered or culturally skewed language and flags contract terms that may be risky or one-sided if signed. Using lightweight NLP models and factual consistency checks, it highlights problematic clauses with visual overlays, providing clause-level feedback without requiring deep legal expertise.

By integrating these components, the system offers a domain-specialized, explainable, and ethically grounded AI framework to enhance the review and preparation of employment agreements—promoting fairness, legal safety, and interpretability in contract law.


Legal document analyzer and summarizer
Generates context-based summaries from legal documents
Trained on real Sri Lankan employment contracts
Visualizes clause types, timelines, and key terms
Interpretable Legal Reasoning Module
Provides step-by-step legal reasoning with source citations
Builds logic trees and argumentation graphs
Enhances auditability and transparency of AI outputs
Knowledge-Driven AI Framework for Legal Contract Understanding and Trust Assessment
Legal Knowledge Graph Builder
Extracts legal entities and relationships from documents
Builds dynamic, auto-updating legal knowledge graphs
Supports cross-referencing across legal modules
Bias & Risk Evaluator
Detects legal-specific bias (e.g., gender, cultural, unfair terms)
Assesses risk levels based on potential legal issues in clause wording
Highlights risky or biased content using visual overlays for clarity

7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

This project focuses exclusively on **Employee Contract Law**, aiming to support legal professionals and HR personnel in drafting, interpreting, and validating employment-related legal documents through AI. Specialized knowledge in employment legislation, statutory obligations, contractual terms (e.g., probation, termination, non-compete, remuneration), and labor-related case law is essential. Understanding how employment law is applied in real-world contracts — including industry-specific norms and regional legal variations in Sri Lanka — is critical to model the system effectively.

Specialized Domain Expertise

- **Legal:** Deep knowledge of Sri Lankan employment and contract law, including the *Shop and Office Employees (Regulation of Employment and Remuneration) Act*, *Wages Boards Ordinance*, *Termination of Employment of Workmen (Special Provisions) Act (TEWA)*, and *Industrial Disputes Act*; familiarity with standard employment-contract structures (e.g., probation, remuneration, leave, termination, confidentiality, IP, dispute resolution) and how courts interpret ambiguous clauses.
- **Governance & Ethics:** Expertise in legal risk, explainability, and responsible AI for high-stakes use; understanding of Sri Lanka's **Personal Data Protection Act** obligations (lawful basis, minimization, purpose limitation).
- **Annotation & Evaluation:** Ability to design clause taxonomies, define reasoning/rationale labels, and set gold-standard guidelines; conduct expert review for accuracy and bias.

Data Requirements include:

- English-language **employment contracts**, policy documents, and HR templates
- Statutory texts from Sri Lankan labor and employment laws
- Case judgments related to employee rights and disputes
- Labeled examples of **bias types** (e.g., gender, cultural) and **risk levels** in contract clauses
- Annotated legal corpora for training models (e.g., LegalBERT)
- Clause libraries with expert validation
- Real-time feeds from legal gazettes and labor court rulings

All data handling adheres to legal and ethical standards, with emphasis on accuracy, transparency, and explainability.

8. Objectives and Novelty

Main Objective			
To develop an explainable, trustworthy, and domain-specialized AI system that assists legal professionals, HR managers, and policymakers in analyzing, drafting, and validating **employee contracts** by leveraging Large Language Models (LLMs), legal knowledge graphs, and bias-aware reasoning — exclusively within the scope of **Sri Lankan Employment Contract Law** and in the **English language**.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
IT22113504 Wicramasinghe D A T N	Enable users to query and understand lengthy employee contracts quickly by extracting relevant clauses and presenting visual summaries.	• Collect and preprocess English-language employee contracts • Implement Retrieval-Augmented Generation (RAG) to enable query-based clause summarization • Design visual dashboards showing clause distributions, timelines, and risks	This component introduces a novel legal document analyzer that generates **user-specific, context-aware summaries** using advanced language models, presents **insights through interactive visual dashboards** (e.g., clause distributions and timelines)

IT22344892 Maxwell L Y	This component aims to help legal professionals identify potential **biases** and **risks** in employment agreement texts using AI. It provides clear, visual feedback on which parts of a contract may be **unfair**, **discriminatory**, or **legally problematic** before signing.	• Detect context-aware bias (e.g., gendered language in job roles or leave clauses) • Assess risk levels based on clause content and potential legal consequences • Provide visual overlays for biased or risky "hotspots" in documents	This component introduces a lightweight yet effective **bias and risk evaluation system** tailored for legal contract texts. Unlike generic bias tools, it offers **clause-level insights** with visual risk indicators, making it practical for contract review and accessible to non-experts.
IT22348784 Sivanuja S	To develop an explainable legal reasoning system for employee contract law that mimics how legal professionals reason through contract clauses. This component addresses the lack of transparency in current AI systems by providing step-by-step, statute-backed explanations, making AI decisions more traceable, defensible, and legally compliant.	• Create chain-of-thought prompts to simulate legal reasoning. • Link contract clauses to relevant laws using a legal knowledge graph. • Retrieve and cite matching statutes or case laws for each clause. • Visualize reasoning using legal logic trees and decision flow diagrams. • Provide contrastive explanations comparing accepted vs. rejected clauses. • Ensure all reasoning paths meet explainability standards (e.g., EU AI Act).	This component introduces an explainable legal reasoning engine that mimics how legal experts interpret contract clauses. Unlike typical black-box models, it provides step-by-step reasoning backed by Sri Lankan statutes and case law. It also includes visual logic trees and contrastive explanations, making AI outputs more transparent, traceable, and usable for legal professionals.

		• Test outputs using real contract examples and legal expert reviews.	
IT22732736 Sharan K	Build a dynamic, auto-updating knowledge graph representing Sri Lankan employee contract law.	• Extract entities and legal relationships from unstructured legal documents. • Construct and update the knowledge graph using Neo4j. • Link contract clauses to relevant laws and precedent cases. • Enable graph-based queries and visual navigation of legal links. • Integrate KG with other modules for reasoning, summarization, and bias detection.	This component introduces the first auto-updating legal knowledge graph for Sri Lankan employee contract law. It extracts and links laws, clauses, and case rulings at a granular level, enabling real-time traceability. Unlike static legal databases, this dynamic graph supports multilingual tags and powers reasoning, summarization, and bias detection across the full AI system.

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
IT22113504 Wicramasinghe D A T N	This component applies core data science skills such as NLP for summarization and clause extraction using models like LegalBERT and GPT-4. Clause segmentation leverages ML classifiers, while data visualization tools (e.g., Plotly, D3.js) are used to present insights like clause distributions and timelines. The end-to-end workflow involves data collection, preprocessing, model evaluation, and interpretability — all central to data science.
IT22344892 Maxwell L Y	This component applies natural language processing techniques to detect bias and assess risk in employment contract clauses — a key area in responsible and ethical AI. It uses classification and scoring methods to flag potentially unfair or legally problematic language, aligning with core data science skills such as supervised learning, text analysis, and model evaluation. Visual overlays help communicate results clearly, ensuring the solution remains interpretable and practically useful for real-world legal applications.
IT22348784 Sivanuja S	This component enables explainable legal reasoning for employee contract clauses by simulating the logical thinking process of legal professionals. It uses prompt chaining and citation-aware retrieval to generate step-by-step outputs backed by Sri Lankan statutes. Visual logic diagrams and contrastive explanations help users understand why a clause is valid or violates the law, improving legal transparency and trust.
IT22732736 Sharan K	This component constructs a live knowledge graph of Sri Lankan employee contract law. It automatically links clauses from contracts to relevant laws and case rulings, allowing AI systems to trace legal meaning and ensure jurisdictional accuracy. The KG supports reasoning, visual explanation, and system-wide traceability, forming the backbone of the entire LegalVision framework.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor				
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

<u>This part is to be filled by the Topic Screening Staff members.</u>

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	

b) Does the proposed topic exhibit novelty?

Yes		No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

***Important**:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment